

Lesson 11

Homework 1 – Movie Night

Three friends sent you the movies they want to watch tonight. They repeated titles and, of course, they do not agree on anything.

1. Start from three lists of titles (repeat some on purpose) and turn each one into a set. Print how many different movies each friend picked.
2. Print every movie that at least one of them wants to watch.
3. Print the movies the three of them want. If there are none, print a message saying so.
4. Print the movies only the first friend wants, using `difference()` with more than one argument. Then get the same result with the `-` operator.
5. Print the movies that are in the first or the second list, but not in both.
6. Ask the user for a title and answer, using `in` / `not in`, if it is already on the big list.
7. With a set comprehension, build a set with the titles longer than 10 characters.
8. The night starts: add one movie with `add()` and three more at the same time with `update()`.
9. Two friends cancel: take one title out with `remove()` and another one with `discard()`. Try both with a title that is NOT in the set and write in a comment what happens in each case.
10. Print the titles one by one with a while loop and `pop()`, until the set is empty. Run it a couple of times, is the order always the same?
11. Answer in a comment: why can we not put the three original lists inside one set?

Homework 2 – The Report Card

The school saves the report cards in tuples, because once a report card is closed nobody should be able to change it.

1. Create a tuple with the name of the student, the grade level, a list with 3 scores and a boolean saying if the student is enrolled.
2. Print the type of (99) and the type of (99,). Explain the difference in a comment.
3. Print the name using indexing, and the first two items using slicing.
4. Try to change the name inside the tuple. Copy the error you get into a comment.
5. Add a new score of 88 to the list that is inside the tuple. Then try to replace the whole list with [100]. Why does one work and the other one does not?
6. Go through the scores with a for loop and print the average.
7. Build a list with at least 5 tuples (name, score). Print the second tuple, and then only the name of that second tuple.
8. Append one more (name, score) to the list and print all of them with a for loop, one per line.
9. Print the last three tuples of the list using slicing with a negative index.
10. Turn that list of tuples into a set. What happens if two students have exactly the same (name, score)? Try the same thing with a list inside and explain the error.
11. Answer in a comment: give one case where a tuple is better than a list, and one case where the list is better.